MIL-P-63165(AR)
25 June 1982
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

PIN
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for pins used in M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-26074	Coatings, Electroless Nickel, Requirements for
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC NUOR

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9299048 Pin

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-8	Tension Testing of Metallic Materials
ASTM E-18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM B-487	Method for Measuring Metal and Oxide Coating Thickness by Microscopic Examination of a Cross Section

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The pins shall comply with all requirements specified on drawing (dwg.) 9299048, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Materials. The contractor shall provide objective evidence (see 6.4) that materials comply with applicable dwgs. and specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

3.3 Protective finish. Protective finished pins (see dwg. 9299048) shall be electroless nickel coated in accordance with MIL-C-26074 (Class 1) with thickness as specified on the dwg. Coating shall be of high quality, fine grained and free of nodules, bare spots, blisters, scratches, flaking, non-adhesion and any other obvious coating defects. Testing shall be performed in accordance with 4.5.3.

3.4 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.5 Workmanship. Pins shall be fabricated and finished in a thoroughly workmanlike manner and shall be free of scratches, burrs and sharp edges. The pins shall be clean and free of chips, dirt, grease, rust, corrosion or any other foreign matter. Cleaning methods shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First Article Inspection (see 4.3)
2. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When the contract requires first article inspection (see 3.4), the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with 4.3.2. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

first article sample shall consist of the following items in the sample quantities as indicated.

Quantity	Part Description	Drawing
6	Completely finished pins	9299048
6	Material specimens adequate to perform applicable yield strength tests	9299048

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and to the tests and inspections specified in the first article inspection table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source as will be used in regular production. When more than one diameter pin is ordered, first article samples shall be required for each different size pin ordered.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63165(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.3.2.1	Pin		1 OF 1		9299048
					NEXT HIGHER ASSEMBLY: M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classfication of Defects and Tests	6(a)			4.4.2.1
	Dimensional characteristics (b)			3.1	
	Hardness			3.1	
	Protective finish thickness (c)			3.3	
	Protective finish quality (c)			3.3	
	Yield strength (d)			3.1	
	Workmanship			3.5	
	NOTES:				
	(a) Six samples required for each size pin ordered				
	(b) For protective finished pins, dimensions apply after protective finishing				
	(c) Applicable to protective finished pins only				
	(d) Six material specimens adequate to perform applicable yield strength tests				

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- 5 -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.2 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots of pins shall comply with the lot formation provisions of MIL-A-48078. In addition, when applicable, a protective finish inspection lot shall consist of items finished at the same time with the same equipment by the same finish process with the same lot of finish material. A protective finish inspection lot shall not exceed 1000 units.

4.4.2 Examinations. See MIL-A-48078 and the Quality Conformance Inspection Tables herein.

(a) Major defects: Unless otherwise specified, inspection for Major defects shall be sampled in accordance with Table I.

(b) Minor defects. Unless otherwise specified, inspection for Minor defects shall be sampled in accordance with Table II.

Table I. Sampling plan for major defects.

LQ = .05 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

Table II. Sampling plan for minor defects.

LQ = .10 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Pin				9299048
					NEXT HIGHER ASSEMBLY
					M753 Proj.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Outside diameter		100%	3.1	Gage
102.	Overall length		100%	3.1	Gage
103.	Improper surface finish on outside diameter		100%	3.1	Gage
104.	Pitch diameter of thread		Table I	3.1	Gage
105.	Minor diameter of thread		Table I	3.1	Gage
106.	Depth of thread (max)		Table I	3.1	Gage
107.	Improper radius (untapped end)		Table I	3.1	Gage
108.	Improper radius (tapped end)		Table I	3.1	Gage
109.	Hardness		100%	3.1	4.5.1
110.	Yield strength	4.4.3.1		3.1	4.5.2
111.	Protective finish thickness (a)	4.4.3.2	-	3.3	4.5.3.1
112.	Protective finish quality (a)		100%	3.3	4.5.3.2
113.	Materials		-	3.2	Visual (see 6.4)
114.	Marking missing or incorrect		Table I	3.1	Visual
Minor:					
201.	Poor workmanship		Table II	3.5	Visual
202.	125 surface finish		Table II	3.1	Visual

NOTES:
(a) Applicable to protective finished pins only.

Source: https://assist.dla.mil -- Downloaded: 2026-02-23T02:21Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Packaging				See 5.0
					NEXT HIGHER ASSEMBLY
					N.A.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements		-	5.1	Visual
102.	Objective evidence (see 3.2) not available for packaging and packing material		-	5.1	Visual
103.	Parts not clean and free of dirt and other foreign material		100%	5.1	Visual
104.	Parts not preserved and individually wrapped or packed in heat sealed barrier bag		100%	5.1	Visual
105.	Improper quantities in container		100%	5.1	Visual
106.	Exterior pack improper or not properly sealed				
107.	Marking of containers or external closures missing, improper or illegible		100%	5.2	Visual
108.	Evidence or poor workmanship		100%	-	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the first Article and Quality Conformance Inspection tables and herein.

4.4.3.1 Yield strength. Two specimens from each heat treat lot of material shall be tested for yield strength in accordance with 4.5.2. The specimens may be of any convenient size or configuration and shall be representative of the complete heat treat lot. If any specimen fails to comply with the yield strength requirements specified on the dwg., the heat treat inspection lot shall be rejected.

4.4.3.2 Protective finish thickness. Specimens or coupons from each protective finish inspection lot finished concurrently with the actual items shall be tested for finish thickness in accordance with 4.5.3.1. The specimens or coupons shall be prepared from the same base material with the same finish and cleaned and processed concurrently with the items being finished. The finish thickness shall be verified on at least two specimens or coupons per protective finish inspection lot. If any specimen or coupon fails to meet the applicable finish thickness requirements, the inspection lot under consideration shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test methods.

4.5.1 Hardness. Rockwell hardness shall be measured in accordance with ASTM E18 or an approved equivalent.

4.5.2 Yield strength. Yield strength shall be tested in accordance with ASTM E8 or an approved equivalent.

4.5.3 Protective finish (see 3.3).

4.5.3.1 Protective finish thickness. Protective finish thickness shall be measured in accordance with ASTM B-487 or an approved equivalent.

4.5.3.2 Protective finish quality. Protective finish quality shall be examined under 4x magnification.

Source: https://assist.dla.mil -- Downloaded: 202610-23T02:21Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract, packaging and packing shall be in accordance with Packaging Data Sheet (PDS) 9299048.

5.2 Marking. Containers and external closures shall be marked with the following:

a. Item nomenclature
b. Part or dwg. no.
c. Used on M753 Projectile
d. Manufacturer
e. Quantity inclosed
f. Pin size
g. Lot no.
h. Contract or purchase order no.
i. Date of acceptance

6. NOTES

6.1 Intended use. Pins covered by this specification are intended for use on Rocket Motor/Bulkhead Assemblies for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. First article inspection, when required (see 3.4).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

Source: https://assist.dla.mil -- Downloaded: 202610-23T02:21Z
Check the source to verify that this is the current version before use.

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence. A statement (certification) supported by test data that each lot of material meets the requirements of the applicable dwgs. and specifications.

6.5 Supersession data. This specification includes the requirements of AR-PD-17, Revision A dated 27 April 1981.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:

Army-AR

Preparing activity

Army-AR

(Project NUOR-A104)

Source: https://assist.dla.mil -- Downloaded: 202610-23T02:21Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAN-Q
Dover, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 202640-23T02:21Z
Check the source to verify that this is the currentversion before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-63165 PIN M753 8 INCH PROJECTILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:21Z
Check the source to verify that this is the current version before use.